



Advanced Time Series Analysis and Forecasting

Seminar 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



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Acronyms used in this material (1/2)

- **AI:** Artificial Intelligence
- **AR:** AutoRegressive (model); in event studies: Abnormal Return
- **BET:** Bucharest Exchange Trading (index)
- **BIC:** Bayesian Information Criterion
- **BNR:** Banca Națională a României (the National Bank of Romania)
- **BV:** Bipower Variation
- **CDS:** Credit Default Swap
- **CLT:** Central Limit Theorem
- **DCC:** Dynamic Conditional Correlation
- **DCC-NL:** DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- **DM:** Diebold–Mariano (test)
- **EODHD:** EOD Historical Data (market-data provider)
- **ETF:** Exchange-Traded Fund
- **FTX:** FTX (cryptocurrency exchange, bankrupt in November 2022)
- **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-MIDAS:** GARCH with a Mixed Data Sampling long-run component (Engle–Ghysels–Sohn 2013)

Acronyms used in this material (2/2)

- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **HARQ**: HAR with realised Quarticity
- ▣ **IQ**: Integrated Quarticity
- ▣ **ML**: Machine Learning
- ▣ **MSE**: Mean Squared Error
- ▣ **OLS**: Ordinary Least Squares
- ▣ **OPG**: Outer Product of the Gradient (scores)
- ▣ **QLIKE**: Quasi-LIKElihood loss
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **QMLE**: Quasi-Maximum Likelihood Estimator
- ▣ **RQ**: Realised Quarticity
- ▣ **RV**: Realised Variance
- ▣ **SCA**: Seasonally and Calendar Adjusted
- ▣ **TQ**: Tripower Quarticity
- ▣ **TSA**: Time Series Analysis
- ▣ **TSRV**: Two-Scales Realised Variance
- ▣ **US**: United States
- ▣ **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ **VR**: Variance Ratio

Today's question and route

- **Question:** how much do we really know about a volatility estimate, and when does a better measurement give a better forecast?
 - ▶ this seminar comes **before** Lecture 8: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: robust standard errors of QML, the noise bias of realised variance, two scales and kernel weights, bipower variation and a jump test by hand
 - ▶ Part A (continued): HAR as an AR(22), attenuation and HARQ, robust losses, one DCC step
 - ▶ Part B: jumps in Ether; HAR for the DAX; HARQ for Ether out of sample; GARCH for EUR/RON with robust standard errors; GARCH-MIDAS for the BET with Romanian industrial production
 - ▶ Part C: did Bitcoin volatility change after the US spot ETFs? An AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	robust standard error of a QML variance; a sandwich by hand	Solved, Proposed	A1
A3, A4	noise bias and optimal sampling; two scales and Parzen weights	Solved, Proposed	A3
A5, A6	RV, BV and the jump test on a toy day; the smallest detectable jump	Solved, Proposed	A5
A7, A8	HAR as a restricted AR(22); attenuation and HARQ	Solved, Proposed	A7
A9, A10	robust and non-robust losses; one DCC and one cDCC step	Solved, Proposed	A9
B1, B2	jumps in Ether; HAR for the DAX	Solved	–
B3	HARQ for Ether out of sample	Proposed	B2, A8
B4, B5	QML for EUR/RON; GARCH-MIDAS for the BET	Solved, Proposed	B4
C1, C2	Bitcoin before and after the spot ETFs; what is wrong in an AI answer?	Proposed	B1–B5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Ether, Bitcoin: realised measures	Binance, one-minute prices	daily	2018–2026
DAX: 5-minute RV and BV	Heber, Lunde, Shephard and Sheppard (2009)	daily	2000–2022
EUR/RON reference rate; BET	BNR; EODHD	daily	2003–2026
Romanian industrial production (SCA)	Eurostat sts_inpr_m	monthly	2000–2026
S&P 500	EODHD	daily	2021–2026

□ Functions in the notebook

► `fit_garch`, `sandwich`, `bipower`, `tripower`, `ht_stat`, `har_X`, `har_forecasts`, `qlike`, `dm_test`,
`fit_garch_midas`

□ No account or key is needed

Prerequisites for Today (1/9): GARCH and quasi-maximum likelihood

- GARCH(1,1) (TSA, Chapter 5): the return shock is a volatility times a standardised shock

$$\varepsilon_t = \sigma_t \eta_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- ▶ ε_t : demeaned return of day t ; σ_t^2 : its conditional variance given the past; $\omega > 0$, $\alpha, \beta \geq 0$: parameters, $\theta = (\omega, \alpha, \beta)$
- ▶ η_t : i.i.d. shock with mean 0 and variance 1; $\kappa_\eta = \mathbb{E}\eta_t^4$: its kurtosis (3 for the Normal distribution)

- QML: maximise the Gaussian log-likelihood even if η_t is not Gaussian

$$\hat{\theta} = \arg \max_{\theta} \sum_{t=1}^T \ell_t(\theta), \quad \ell_t = -\frac{1}{2} \left(\ln \sigma_t^2 + \frac{\varepsilon_t^2}{\sigma_t^2} \right)$$

- ▶ T : number of days; ℓ_t : Gaussian log-density of day t without its constant; $\hat{\theta}$ is consistent if the variance equation is right

Prerequisites for Today (2/9): the sandwich

- The variance of the QML estimator is a “sandwich” of two matrices

$$\text{Var}(\hat{\theta}) \approx \frac{1}{T} A^{-1} B A^{-1}, \quad A = -\frac{1}{T} \sum_t \frac{\partial^2 \ell_t}{\partial \theta \partial \theta'}, \quad B = \frac{1}{T} \sum_t s_t s_t'$$

- ▶ $s_t = \partial \ell_t / \partial \theta$: the score of day t ; A : minus the average Hessian; B : the average outer product of the scores
 - ▶ Hessian only: $\frac{1}{T} A^{-1}$; outer product of the scores (OPG) only: $\frac{1}{T} B^{-1}$; both are valid only if the likelihood is the true one ($A = B$)
- For GARCH the sandwich is $(\kappa_\eta - 1)/2$ times the Hessian-only variance
 - ▶ standard errors from the sandwich: Bollerslev–Wooldridge; with fat tails ($\kappa_\eta > 3$) the Hessian ones are too small

Prerequisites for Today (3/9): GARCH-MIDAS

- GARCH-MIDAS: the daily variance is a monthly level τ_t times a daily GARCH factor $g_{i,t}$ with mean 1

$$r_{i,t} = \mu + \sqrt{\tau_t g_{i,t}} \eta_{i,t}, \quad \ln \tau_t = m + \theta \sum_{k=1}^K \varphi_k(w) X_{t-k}$$

- ▶ $r_{i,t}$: return of day i of month t ; μ : mean return; X_{t-k} : a monthly variable k months ago; K : number of monthly lags
- ▶ m : intercept; θ : effect of X on the long-run variance; $\varphi_k(w) \propto (1 - k/K)^{w-1}$: decaying beta weights that sum to 1 ($w = 1$: equal weights)
- Variance ratio: the share of the variation of log volatility due to the long-run component

$$\text{VR} = \frac{\text{Var}(\ln \tau_t)}{\text{Var}(\ln(\tau_t g_{i,t}))} \in [0, 1]$$

- ▶ $\text{BIC} = -2 \ln L + k \ln T$, with L the maximised likelihood and k the number of parameters: lower is better

Prerequisites for Today (4/9): realised measures and noise

- Realised variance from n intraday returns r_i of a day, and its precision

$$RV = \sum_{i=1}^n r_i^2, \quad \text{Var}(RV - IV) \approx \frac{2IQ}{n}, \quad \widehat{IQ} = RQ = \frac{n}{3} \sum_{i=1}^n r_i^4$$

- ▶ IV: integrated variance, the true variance of the day; without noise $RV \rightarrow QV$, the quadratic variation = IV plus the squared jumps
- ▶ IQ: integrated quarticity (the daily average of σ^4); RQ: realised quarticity, its estimate

- Noise: the observed log price is $Y = X + u$

$$\mathbb{E}RV = IV + 2n\omega^2, \quad n^* \approx \left(\frac{IQ}{4\omega^4} \right)^{1/3}$$

- ▶ X : efficient log price; u : i.i.d. noise with mean 0 and variance ω^2 ; the bias $2n\omega^2$ grows with the sampling frequency
- ▶ n^* : the number of returns that minimises the approximate MSE $2IQ/n + (2n\omega^2)^2$

Prerequisites for Today (5/9): two scales and the realised kernel

- Two-scales RV: the noise bias of a sparse RV is estimated from the dense RV and subtracted

$$\text{TSRV} = \overline{\text{RV}}^{(K)} - \frac{\bar{n}}{n} \text{RV}^{(n)}, \quad \bar{n} = \frac{n - K + 1}{K}$$

- ▶ $\text{RV}^{(n)}$: RV from all n returns; $\overline{\text{RV}}^{(K)}$: the average RV over K offset sparse grids, each with about $\bar{n}$ returns

- Realised kernel: RV plus weighted autocovariances, which cancel the noise bias

$$\text{RK} = \gamma_0 + 2 \sum_{h=1}^H k\left(\frac{h}{H+1}\right) \gamma_h, \quad \gamma_h = \sum_i r_i r_{i-h}$$

- ▶ γ_h : realised autocovariance at lag h ; H : bandwidth (number of lags); $k(\cdot)$: weight function
- ▶ Parzen weights: $k(x) = 1 - 6x^2 + 6x^3$ for $x \leq \frac{1}{2}$ and $2(1-x)^3$ for $\frac{1}{2} < x \leq 1$

Prerequisites for Today (6/9): bipower variation and the jump test

- Bipower variation estimates IV even with jumps; tripower quarticity estimates IQ even with jumps

$$\text{BV} = \mu_1^{-2} \frac{n}{n-1} \sum_{i=2}^n |r_i| |r_{i-1}|, \quad \text{TQ} = n \mu_{4/3}^{-3} \frac{n}{n-2} \sum_{i=3}^n \prod_{j=0}^2 |r_{i-j}|^{4/3}$$

- ▶ $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$ for $Z \sim N(0, 1)$: constants that make BV and TQ consistent

- Huang–Tauchen test: the relative jump contribution $(\text{RV} - \text{BV})/\text{RV}$, studentised

$$z = \frac{(\text{RV} - \text{BV})/\text{RV}}{\sqrt{\vartheta n^{-1} \max(1, \text{TQ}/\text{BV}^2)}} \approx N(0, 1), \quad \vartheta = \frac{\pi^2}{4} + \pi - 5 \approx 0.609$$

- ▶ reject “no jump on this day” if $z > z_{1-\alpha}$, the $1 - \alpha$ quantile of $N(0, 1)$; at $\alpha = 0.1\%$, $z_{0.999} = 3.09$

Prerequisites for Today (7/9): HAR and HARQ

- HAR: tomorrow's RV regressed on the daily, weekly and monthly averages of past RV

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + u_{t+1}$$

- ▶ $RV_t^{(w)}$, $RV_t^{(m)}$: means of RV over the last 5 and 22 days; $\beta_d + \beta_w + \beta_m$: persistence; u_{t+1} : error
- ▶ OLS with Newey–West (HAC) standard errors, robust to heteroskedasticity and autocorrelation

- Errors in variables: if $RV_t = IV_t + e_t$ and $IV_{t+1} = c + \phi IV_t + v_{t+1}$, the OLS slope is attenuated

$$\text{plim } \hat{\phi} = \phi\lambda, \quad \lambda = \frac{\text{Var}(IV)}{\text{Var}(IV) + \text{Var}(e)}$$

- ▶ e_t : measurement error of RV; λ : reliability ratio, the share of signal in the variance of RV
- ▶ HARQ: β_d is replaced by $\beta_d + \beta_{dQ} \sqrt{RQ_t}$, so yesterday's weight falls on imprecise days ($\beta_{dQ} < 0$)

Prerequisites for Today (8/9): evaluating volatility forecasts

- Two robust losses for a forecast h against a proxy $\hat{\sigma}^2$ (Patton 2011)

$$\text{QLIKE} = \frac{\hat{\sigma}^2}{h} - \ln \frac{\hat{\sigma}^2}{h} - 1, \quad \text{MSE} = (\hat{\sigma}^2 - h)^2$$

- ▶ proxy: an unbiased but noisy measure of the true variance (squared return, RV); robust: the ranking of forecasts is the one the true variance would give
- For $z \sim N(0, 1)$: $\mathbb{E} \ln z^2 = \psi(\frac{1}{2}) + \ln 2 \approx -1.27$ and $\text{Var} \ln z^2 = \pi^2/2$
 - ▶ ψ : the digamma function, the derivative of $\ln \Gamma$; these moments are needed for the MSE-log loss $(\ln \hat{\sigma}^2 - \ln h)^2$
- Insanity filter: a forecast outside the in-sample range of RV is replaced by the in-sample mean
- Diebold–Mariano (DM): t -statistic of the mean loss differential with a HAC variance; $|DM| > 1.96$: significant difference at 5%

Prerequisites for Today (9/9): DCC and cDCC

- DCC: a GARCH-type recursion for a quasi-correlation matrix Q_t of the standardised residuals

$$Q_t = (1 - a - b)S + a z_{t-1} z'_{t-1} + b Q_{t-1}, \quad \rho_{12,t} = \frac{q_{12,t}}{\sqrt{q_{11,t} q_{22,t}}}$$

- ▶ z_t : vector of returns divided by their own GARCH volatilities; S : sample correlation matrix of z_t (the target)
- ▶ a : reaction to yesterday's co-movement; b : persistence; $q_{ij,t}$: elements of Q_t ; $\rho_{12,t}$: the conditional correlation of assets 1 and 2

- cDCC: z_{t-1} is replaced by $z_{t-1}^* = \text{diag}(Q_{t-1})^{1/2} z_{t-1}$

- ▶ $\text{diag}(Q_{t-1})^{1/2}$: diagonal matrix with $\sqrt{q_{ii,t-1}}$; the correction makes the estimator of S consistent

A1: the QML variance and its robust standard error [Solved]

Task

- ▣ BET daily returns (%), demeaned, 2016–2026, $T = 2\,674$; model $r_t = \sigma \eta_t$ with η_t i.i.d., mean 0, variance 1, kurtosis κ .
- ▣ 1. Derive the QMLE $\hat{\sigma}^2$ and its Hessian-only variance under normality.
- ▣ 2. Derive the sandwich variance and compute both standard errors from $\hat{\sigma}^2 = 0.993$ and $\hat{\kappa} = 20.79$.
- ▣ 3. State by how much the Hessian understates the uncertainty.
- ▣ Report: one estimator, two standard errors, one ratio, one sentence.

Solution

- ▣ 1. $\ell_t = -\frac{1}{2}(\ln \sigma^2 + r_t^2/\sigma^2)$; $\hat{\sigma}^2 = \frac{1}{T} \sum r_t^2$; $A = \frac{1}{2\sigma^4}$, so the Hessian-only variance is $2\sigma^4/T$
- ▣ 2. $s_t = \frac{r_t^2 - \sigma^2}{2\sigma^4}$, $B = \frac{(\kappa-1)\sigma^4}{4\sigma^8}$; $A^{-1}BA^{-1} = (\kappa-1)\sigma^4$: s.e. 0.0272 (Hessian) and 0.0854 (sandwich)
- ▣ 3. Ratio $\sqrt{(\hat{\kappa}-1)/2} = 3.15$: with the BET's fat tails, Hessian intervals are about three times too narrow

A2: a sandwich by hand [Proposed]

Solution discussed in class

Task

- ▣ GARCH(1,1) by QML, $T = 4000$; per-observation averages for (α, β) : $A = \begin{pmatrix} 50 & 40 \\ 40 & 36 \end{pmatrix}$, $B = \begin{pmatrix} 120 & 90 \\ 90 & 72 \end{pmatrix}$. Model: A1.
- ▣ 1. Compute A^{-1} and the Hessian-only standard errors $\sqrt{\text{diag}(A^{-1})/T}$.
- ▣ 2. Compute $A^{-1} B A^{-1}$ and the sandwich standard errors.
- ▣ 3. Compute the OPG standard errors and say which of the three you would report.
- ▣ Report: three pairs of standard errors, one sentence.

A3: the noise bias and the optimal sampling frequency [Solved]

Task

- ▣ A 6.5-hour day, constant volatility with $IV = IQ = 1$ ($\%^2$, $\%^4$), i.i.d. noise with $\omega^2 = 1.6 \times 10^{-5}$.
- ▣ 1. Compute $\mathbb{E}RV$ at 1 second ($n = 23\,400$), 1 minute ($n = 390$) and 5 minutes ($n = 78$).
- ▣ 2. Compute the approximate $MSE\ 2IQ/n + (2n\omega^2)^2$ at the three frequencies.
- ▣ 3. Derive n^* and convert it into seconds.
- ▣ Report: three expectations, three MSE values, one optimal interval.

Solution

- ▣ 1. $\mathbb{E}RV = 1 + 2n\omega^2$: 1.7488, 1.0125, 1.0025
- ▣ 2. MSE: 0.5608 (bias dominates), 0.0053, 0.0256 (variance dominates)
- ▣ 3. $\frac{d}{dn} \left[\frac{2}{n} + 4n^2\omega^4 \right] = 0 \Rightarrow n^* = (4\omega^4)^{-1/3} = 992$: one return every 23.6 seconds

A4: two scales and kernel weights [Proposed]

Solution discussed in class

Task

- ▣ One day: $RV^{(n)} = 1.75$ at one second ($n = 23\,400$), average five-minute RV over the $K = 300$ offset grids $\overline{RV}^{(K)} = 1.005$. Model: A3.
- ▣ 1. Compute $\bar{n}$, TSRV and its small-sample version $TSRV/(1 - \bar{n}/n)$.
- ▣ 2. Estimate ω^2 by $RV^{(n)}/(2n)$ and say why it is biased upwards.
- ▣ 3. Compute the Parzen weights $k(h/5)$ for $h = 1, \dots, 4$ ($H = 4$).
- ▣ Report: three numbers, one estimate and a sentence, four weights.

A5: RV, BV and the jump test on a toy day [Solved]

Task

- ▣ Eight returns (%):
0.10, -0.12, 0.08, 1.20, -0.09, 0.11, -0.07, 0.10.
- ▣ 1. Compute RV and BV; compute RV without the fourth return.
- ▣ 2. Compute TQ, the ratio $(RV - BV)/RV$ and the Huang–Tauchen z .
- ▣ 3. Decide at 0.1% and explain why $n = 8$ makes the test weak.
- ▣ Report: three numbers, three numbers, one decision and one sentence.

Solution

- ▣ 1. $RV = 1.5059$; $BV = 0.4492$; without the jump $RV = 0.0659$: the jump dominates RV, BV is far less affected
- ▣ 2. $TQ = 0.1360$, $TQ/BV^2 < 1$ so $\max = 1$; ratio 0.7017;
 $z = 0.7017 / \sqrt{0.609/8} = 2.54$
- ▣ 3. $z < 3.09$: no rejection at 0.1% ($p = 0.0055$); with eight returns BV still carries the jump through two products, and ϑ/n is large

A6: the smallest detectable jump [Proposed]

Solution discussed in class

Task

- ▣ Constant volatility, $IV = 1$, so $BV \approx 1$, $TQ/BV^2 \approx 1$; one jump of size J (in units of $\sqrt{TV}$), so $RV \approx 1 + J^2$. Model: A5.
- ▣ 1. Write z as a function of J and n .
- ▣ 2. Solve $z = 3.09$ for J with $n = 78$ and $n = 390$.
- ▣ 3. Compare with the power curve of the lecture and say what limits finer sampling.
- ▣ Report: one formula, two thresholds, one sentence.

A7: HAR as a restricted AR(22) [Solved]

Task

- ▣ HAR with $\beta_0 = 0.10$, $\beta_d = 0.30$, $\beta_w = 0.40$, $\beta_m = 0.20$.
- ▣ 1. Write the implied AR(22) coefficients ϕ_j on RV_{t+1-j} .
- ▣ 2. Compute their sum and the unconditional mean of RV.
- ▣ 3. Count the restrictions HAR imposes on the AR(22).
- ▣ Report: three distinct coefficients, two numbers, one count.

Solution

- ▣ 1. $\phi_1 = \beta_d + \frac{\beta_w}{5} + \frac{\beta_m}{22} = 0.3891$;
 $\phi_{2..5} = \frac{\beta_w}{5} + \frac{\beta_m}{22} = 0.0891$; $\phi_{6..22} = \frac{\beta_m}{22} = 0.0091$
- ▣ 2. $\sum \phi_j = \beta_d + \beta_w + \beta_m = 0.90$; mean $\beta_0 / (1 - \sum \phi_j) = 1.00$
- ▣ 3. 23 coefficients reduced to 4: 19 linear restrictions (three steps)



A8: attenuation and HARQ [Proposed]

Solution discussed in class

Task

- ▣ $IV_{t+1} = c + 0.95 IV_t + v_{t+1}$, $\text{Var}(IV) = 0.8$; $RV_t = IV_t + e_t$ with $\mathbb{E}\text{Var}(e_t) = 0.2$; $n = 78$. Model: A7.
- ▣ 1. Compute λ and the probability limit of the OLS slope of RV_{t+1} on RV_t .
- ▣ 2. Compute λ_t and the slope on a quiet day ($IQ_t = 0.5$) and on a turbulent day ($IQ_t = 5$), with $\text{Var}(e_t) = 2IQ_t/n$.
- ▣ 3. Explain the sign of β_{dQ} in HARQ.
- ▣ Report: two numbers, four numbers, one sentence.

A9: robust and non-robust losses against the squared return [Solved]

Task

- ▣ $r_t = \sigma_t z_t$, $z_t \sim N(0, 1)$, $\sigma_t^2 = 1$; proxy $\hat{\sigma}_t^2 = r_t^2$; forecasts $h = 1$ (true) and $h = 0.6$ (biased).
- ▣ 1. Find the forecast that minimises $\mathbb{E}(\ln r_t^2 - \ln h)^2$.
- ▣ 2. Compute the expected MSE-log and QLIKE losses of both forecasts.
- ▣ 3. Say which loss ranks the forecasts correctly.
- ▣ Report: one number, four numbers, one sentence.

Solution

- ▣ 1. $h^* = \exp(\mathbb{E} \ln z^2) = \exp(-1.2704) = 0.2807$: far below the true variance
- ▣ 2. MSE-log: $\frac{\pi^2}{2} + (\mathbb{E} \ln z^2 - \ln h)^2$: 6.5486 ($h = 1$) and 5.5117 ($h = 0.6$); QLIKE: $\frac{1}{h} + 1.27 + \ln h - 1$: 1.2704 and 1.4262
- ▣ 3. MSE-log prefers the biased forecast; QLIKE (and MSE: 2 against 2.16) prefer the true one

A10: one DCC and one cDCC update [Proposed]

Solution discussed in class

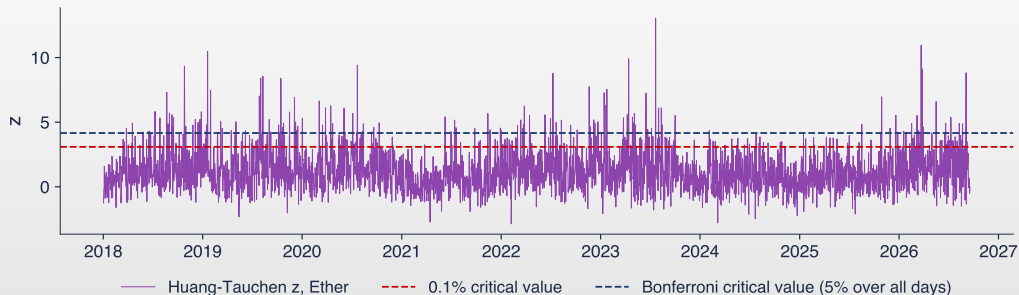
Task

- ▣ $S = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, $a = 0.05$, $b = 0.93$, $Q_{t-1} = \begin{pmatrix} 1.2 & 0.7 \\ 0.7 & 0.9 \end{pmatrix}$, $z_{t-1} = (1.5, -0.5)'$. Model: A9.
- ▣ 1. Compute Q_t and $\rho_{12,t}$ with DCC.
- ▣ 2. Compute z_{t-1}^* , Q_t and $\rho_{12,t}$ with cDCC.
- ▣ 3. Compare with $\rho_{12,t-1}$ and explain the direction of the change.
- ▣ Report: a matrix and a correlation, a vector, a matrix and a correlation, one sentence.

B1: jumps in Ether [Solved]

- ▣ **Question:** how often does Ether jump, and how much of its variation do jumps explain?
- ▣ **Inputs:** our daily RV, BV and TQ from 5-minute returns ($n = 288$ a UTC day), 2018–2026, $T = 3\,165$
- ▣ **Tasks**
 1. Compute the Huang–Tauchen statistic for every day and count the days above the 0.1% critical value.
 2. Compare with the number of false rejections expected under the null, and repeat with a Bonferroni threshold for 5% over all days.
 3. Compute the share of total RV due to the significant jumps.
 4. Interpretation: are jumps frequent, large, or both?
- ▣ **Report:** two counts, one expected number, one share, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]

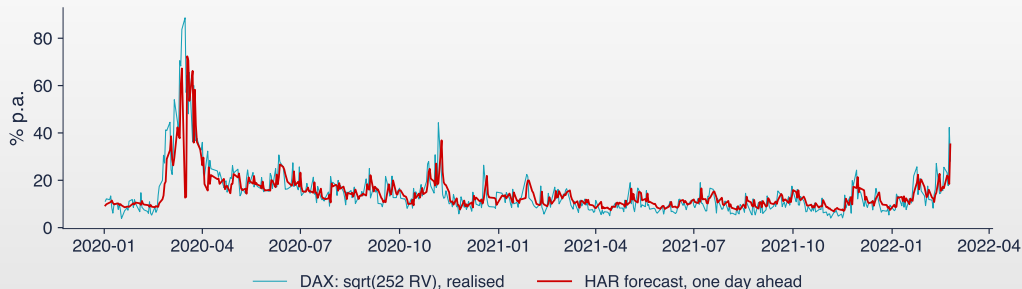


- ▣ At 0.1%: 355 jump days (11.2%) where 3.2 false rejections are expected; Bonferroni threshold $z > 4.16$: still 137 days
- ▣ Largest statistics: 22.07.2023 ($z = 13.1$), 23.03.2026 (10.9), 19.01.2019 (10.5); jump share of total RV 2.3%
- ▣ Interpretation: rejections are far too frequent for the null (mean $z = 1.19$), but jumps carry a small share of variation: frequent small “jumps” partly reflect intraday seasonality and zero returns, a few days are genuine large jumps

B2: HAR for the DAX [Solved]

- **Question:** how persistent is DAX realised variance, and does HAR beat the random walk out of sample?
- **Inputs:** Oxford-Man 5-minute RV of the DAX ($\%^2$), 2000–2022, $T = 5\,589$; rolling window of 1 000 days with the insanity filter
- **Tasks**
 1. Estimate HAR by OLS on the full sample and report Newey–West standard errors (5 lags), R^2 and the persistence.
 2. Produce one-day-ahead HAR, log-HAR and random-walk (RV_t) forecasts and compute the average QLIKE.
 3. Test HAR against each alternative with Diebold–Mariano.
 4. Interpretation: which horizon component matters most for the DAX?
- **Report:** four coefficients with s.e., two statistics, three average losses, two DM statistics, one sentence
- **Notebook:** section B2

B2: solution [Solved]



- ▣ $\hat{\beta}_d = 0.414$ (0.069), $\hat{\beta}_w = 0.279$ (0.107), $\hat{\beta}_m = 0.210$ (0.060), $\hat{\beta}_o = 0.151$; $R^2 = 0.561$; persistence 0.903
- ▣ Out of sample from 22.01.2004 (4 589 days): QLIKE HAR 0.1863, random walk 0.2280 (DM 2.75), log-HAR 0.1597 (DM -2.09)
- ▣ Interpretation: the daily component carries the largest weight for the DAX, the monthly one keeps the forecast anchored; HAR beats the random walk clearly and is beaten by its log version

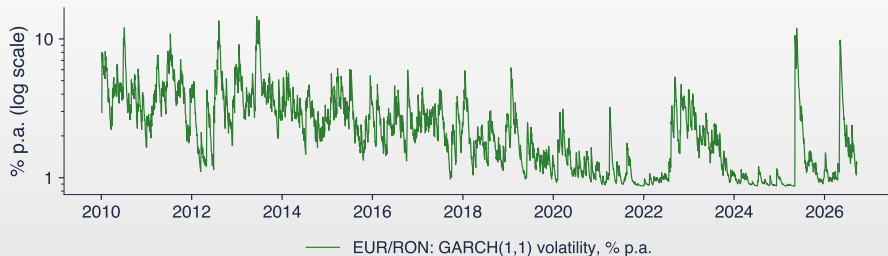
B3: HARQ for Ether out of sample [Proposed]

- **Question:** does the HARQ gain of Bollerslev, Patton and Quaadvlieg (2016) hold for Ether?
- **Inputs:** Ether 5-minute RV and RQ, 2018–2026; rolling window of 1 000 days; model: B2 and A8
- **Tasks**
 1. Produce HAR and HARQ forecasts with the insanity filter and compute QLIKE and MSE ratios HARQ/HAR with DM statistics.
 2. Count the raw HARQ forecasts outside the in-sample range and the negative ones.
 3. Replace the filter by a floor at the smallest in-sample RV and repeat.
 4. Interpretation: is the gain robust to the treatment of implausible forecasts?
- **Report:** four ratios with DM statistics, two counts, one sentence
- **Notebook:** section B3

B4: QML for EUR/RON [Solved]

- **Question:** how persistent is EUR/RON volatility, and how much does the choice of standard errors matter?
- **Inputs:** daily log returns of the BNR reference rate (%), demeaned, 2010–2026, $T = 4\,207$
- **Tasks**
 1. Estimate GARCH(1,1) by Gaussian QML; report Hessian and Bollerslev–Wooldridge standard errors.
 2. Compute the kurtosis of the standardised residuals and compare the ratio of standard errors with $\sqrt{(\hat{\kappa} - 1)/2}$.
 3. Estimate GARCH- t by ML and compare the log-likelihoods.
 4. Interpretation: what does the persistence say about a managed float?
- **Report:** a table of three parameters with two s.e. each, one kurtosis, one log-likelihood gain, one sentence
- **Notebook:** section B4

B4: solution [Solved]



- ▣ $\hat{\omega} = 0.00049$ (0.00004; 0.00027), $\hat{\alpha} = 0.166$ (0.015; 0.040), $\hat{\beta} = 0.833$ (0.012; 0.027); in brackets: Hessian; sandwich
- ▣ Kurtosis of standardised residuals 61.8 (raw returns 15.4); sandwich/Hessian ratio for α 2.71; GARCH- t : $\hat{\nu} = 7.9$, log-likelihood higher by 861.0
- ▣ Interpretation: persistence 0.9998, almost integrated: long calm periods with rare bursts (2012–2013, late 2022, 2025, 2026); the extreme kurtosis even after GARCH filtering shows that one variance equation cannot describe a managed exchange rate

B5: GARCH-MIDAS for the BET [Proposed]

- **Question:** does Romanian industrial production drive the long-run volatility of the BET?
- **Inputs:** BET daily returns from 2003; Eurostat industrial production (B–D, seasonally and calendar adjusted), monthly log growth; $K = 24$ lags; model: B4
- **Tasks**
 1. Estimate GARCH-MIDAS with $\log \tau_t$ and restricted beta weights by QML; report $\hat{\theta}$ and $\hat{w}$ with sandwich standard errors.
 2. Compute the variance ratio.
 3. Compare with GARCH(1,1) on the same days by BIC.
 4. Interpretation: is the macro channel visible in Romanian stock volatility?
- **Report:** two estimates with s.e., one ratio, two BIC values, one sentence
- **Notebook:** section B5

C1: Bitcoin before and after the spot ETFs [Proposed]

- ▣ **Question:** did Bitcoin volatility become more “equity-like” after the US spot ETFs started trading on 11 January 2024?
- ▣ **Inputs:** our Bitcoin realised measures, 2021–2026; S&P 500 daily returns; model: B1–B3
- ▣ **Tasks**
 1. Write a five-line pre-registration: hypothesis, sample split, statistics, tests, what would falsify the claim.
 2. For 2021–2023 and from 11 January 2024, compute annualised volatility (from the realised kernel), the share of jump days, the jump share of RV, the HAR persistence and the correlation with the S&P 500.
 3. Say which differences you could attribute to the ETFs and which to other events.
 4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- ▣ **Report:** a five-line plan, a table of two rows, a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise volatility modelling with high-frequency data. The answer:
- ▣ (a) Gaussian QML is inconsistent when returns are fat-tailed; use Student-t ML instead.
- ▣ (b) Sampling prices as often as possible always makes realised variance more precise.
- ▣ (c) On a day without jumps, bipower variation equals realised variance.
- ▣ (d) HAR is a long-memory model.
- ▣ (e) MSE on log variances is a robust loss for comparing forecasts against squared returns.
- ▣ (f) In DCC the target S is the unconditional correlation of the standardised residuals, so the estimator is consistent.
- ▣ (g) In Realized GARCH the realised measure may be a biased and noisy signal of the conditional variance.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, a number from today's tasks (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ QML needs only the variance equation; its standard errors need the kurtosis: use the sandwich
- ▣ Realised variance trades variance against noise bias; kernels and two scales use all the data
- ▣ Jump tests reject often in real data; jumps explain little of the variation
- ▣ HAR is a restricted AR(22); HARQ lets the precision of RV set yesterday's weight
- ▣ Rank volatility forecasts with QLIKE or MSE; an AI answer is a draft to audit

After the seminar

- ▣ Lecture 8 develops each topic of today
 - ▶ QML theory, component GARCH and GARCH-MIDAS, the CLT of RV, noise and kernels, jump tests
 - ▶ HAR and HARQ, Realized GARCH, robust losses, DCC, cDCC and DCC-NL
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: the realised volatility of crypto-assets around a market-structure change
- ▣ Reading: [Bollerslev and Wooldridge \(1992\)](#); [Barndorff-Nielsen and Shephard \(2002\)](#); [Corsi \(2009\)](#); [Bollerslev, Patton and Quaedvlieg \(2016\)](#); [Patton \(2011\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References (1/2)

- ▣ Aielli, G. P. (2013). Dynamic conditional correlation: On properties and estimation. *Journal of Business & Economic Statistics*, 31(3), 282–299. doi:10.1080/07350015.2013.771027
- ▣ Bandi, F. M., & Russell, J. R. (2008). Microstructure noise, realized variance, and optimal sampling. *The Review of Economic Studies*, 75(2), 339–369. doi:10.1111/j.1467-937X.2008.00474.x
- ▣ Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536. doi:10.3982/ECTA6495
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280. doi:10.1111/1467-9868.00336
- ▣ Barndorff-Nielsen, O. E., & Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37. doi:10.1093/jjfinec/nbh001
- ▣ Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18. doi:10.1016/j.jeconom.2015.10.007
- ▣ Bollerslev, T., & Wooldridge, J. M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172. doi:10.1080/07474939208800229
- ▣ Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001

References (2/2)

- ▣ Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- ▣ Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350. doi:10.1198/073500102288618487
- ▣ Engle, R. F., Ghysels, E., & Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *The Review of Economics and Statistics*, 95(3), 776–797. doi:10.1162/REST_a_00300
- ▣ Heber, G., Lunde, A., Shephard, N., & Sheppard, K. (2009). *Oxford-Man Institute's realized library*, version 0.3. Oxford-Man Institute, University of Oxford (discontinued in 2022; copy of 28 February 2022 preserved by the Internet Archive). web.archive.org
- ▣ Huang, X., & Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499. doi:10.1093/jjfinec/nbi025
- ▣ Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- ▣ Zhang, L., Mykland, P. A., & Aït-Sahalia, Y. (2005). A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411. doi:10.1198/016214505000000169